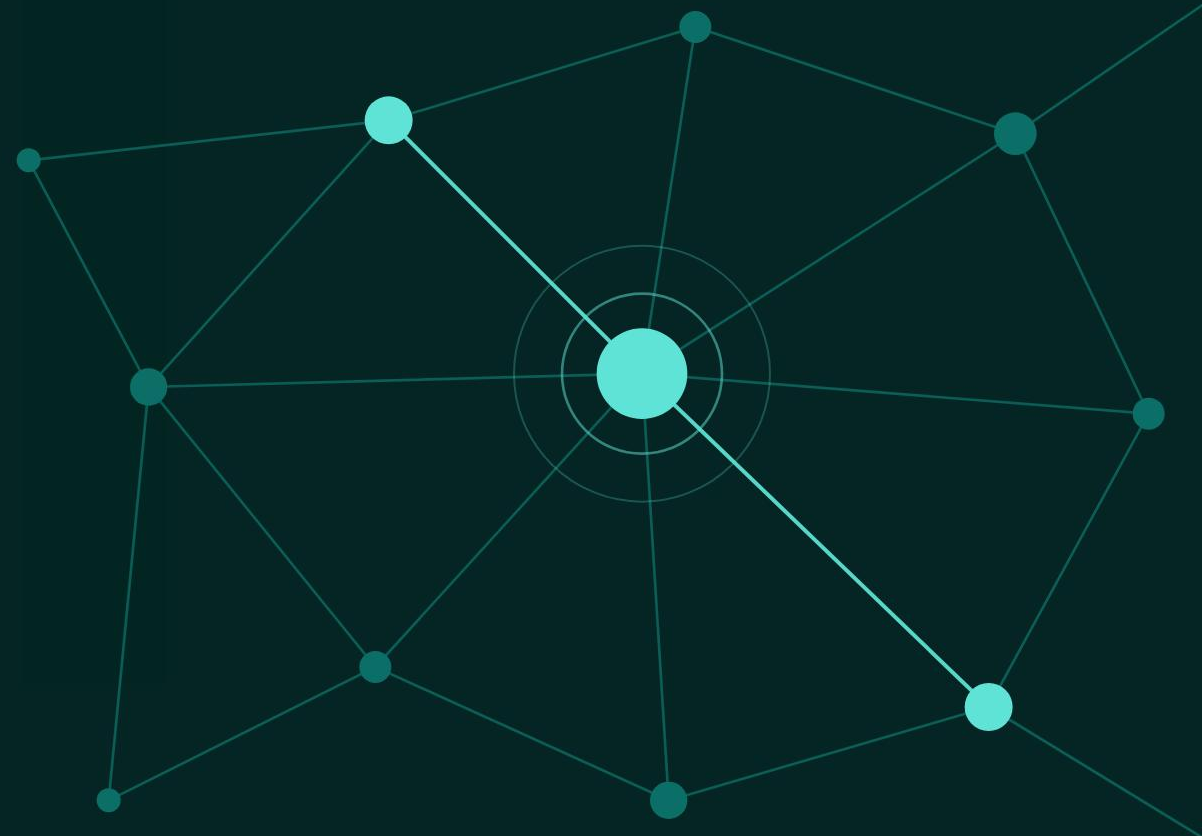


AI COLLECTIVE × QWEN

Hack for Humanity

Learning in the Age of AI

Julia Muiruri • @juliamuiruri4



“

What would you rather have? A tool that
thinks for you, or a tool that makes you
think?

Advait Sarkar · Senior Researcher, Microsoft Research

THE PROBLEM

Is AI killing critical thinking?

Kosmyna et al., MIT Media Lab, 2025 (preprint)

83%

of students who wrote with ChatGPT couldn't quote their own essay, minutes later.

Wrote with ChatGPT

15 of 18



Wrote on their own

2 of 18



COGNITIVE OFFLOADING

It has a cognitive cost

Creativity



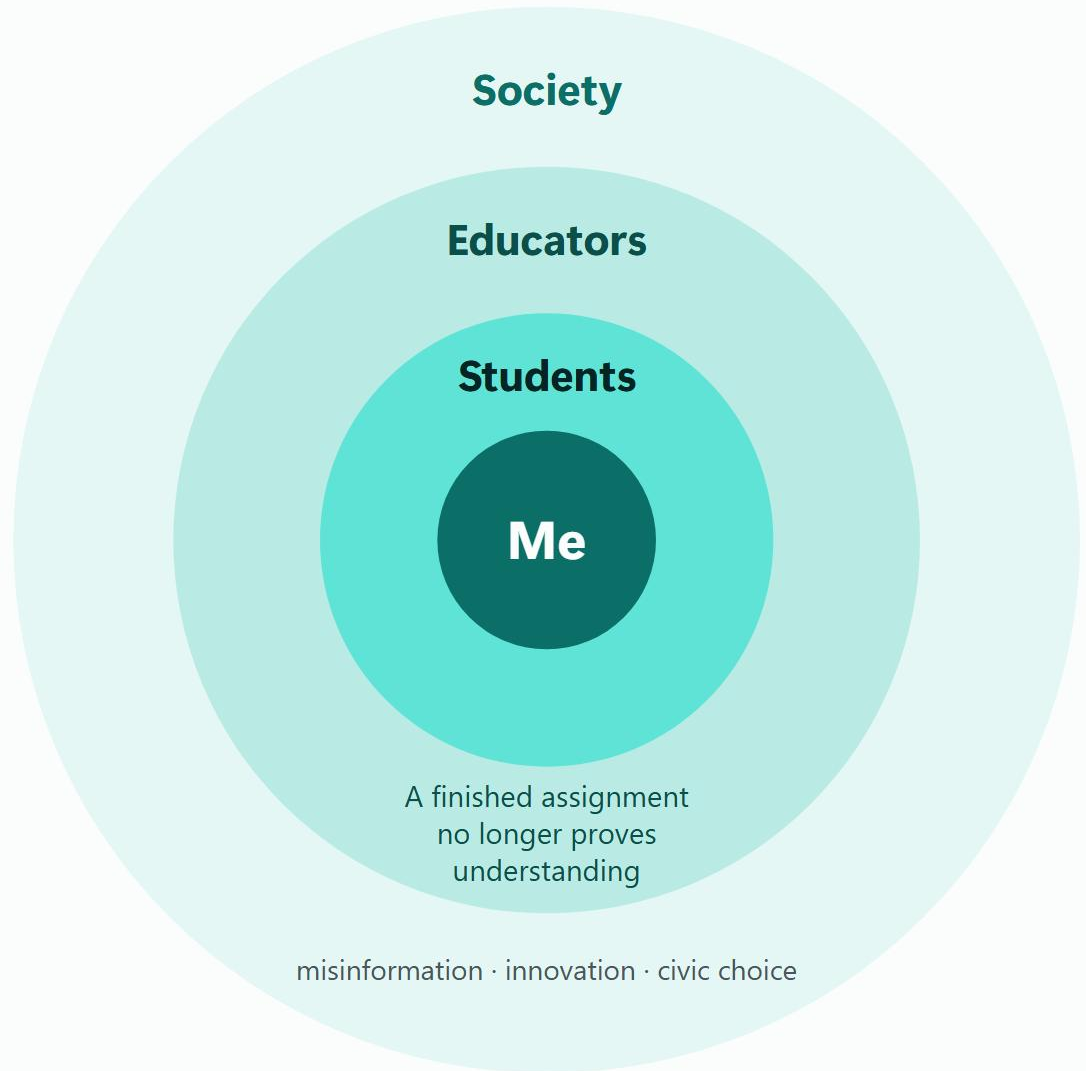
Critical thinking



Memory



What happens to a society that stops thinking?



Not all effort is **bad**

AI should remove the **wrong** effort, never the right one.

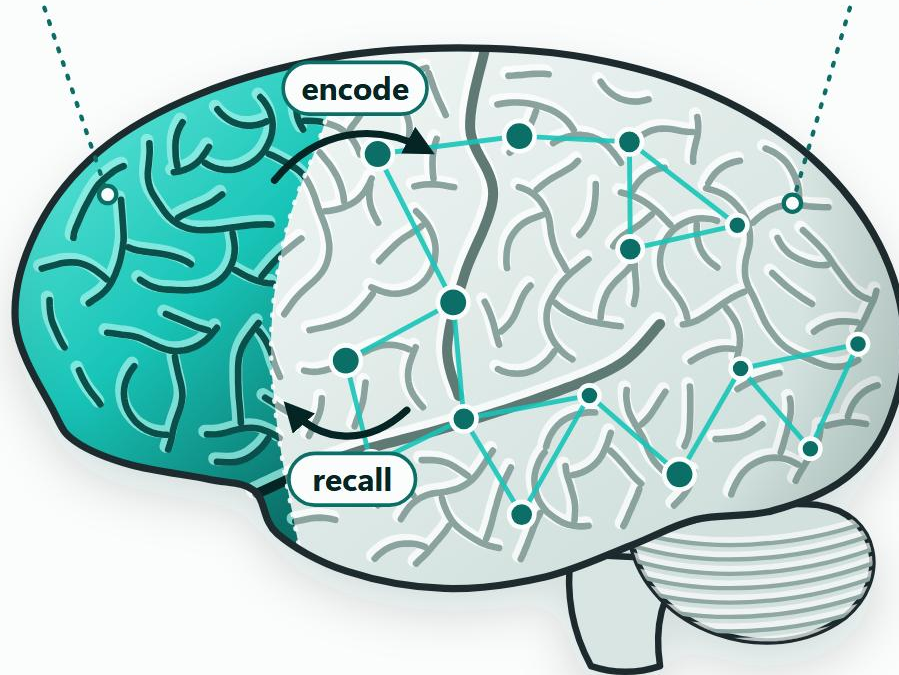
- Extraneous: wasted effort
- Germane: learning

Working memory

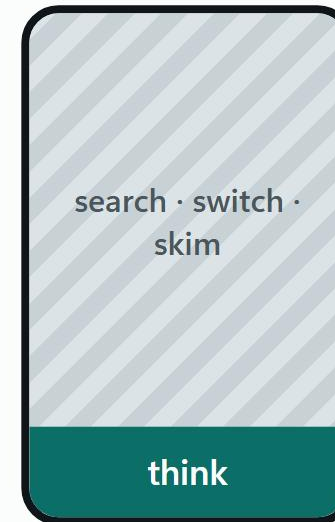
small · about 4 things at once

Long-term memory

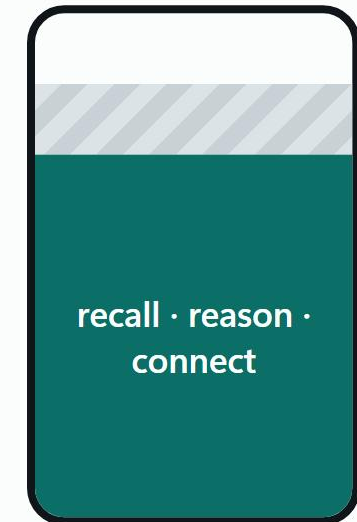
vast · everything you've learned



● Inside your working memory



Overloaded



Cut extraneous. Protect germane.



StudySynk

PROVEN METHODS FIRST. AI SECOND.

AI does the scaffolding.
You do the thinking.

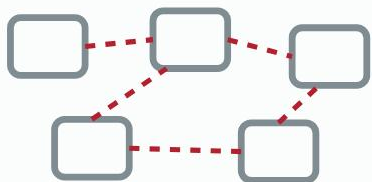
AI

- Finds sources
- Builds maps
- Writes cards
- Keeps time



YOU

- Reason
- Connect
- Recall
- Focus



PROBLEM

The switch tax

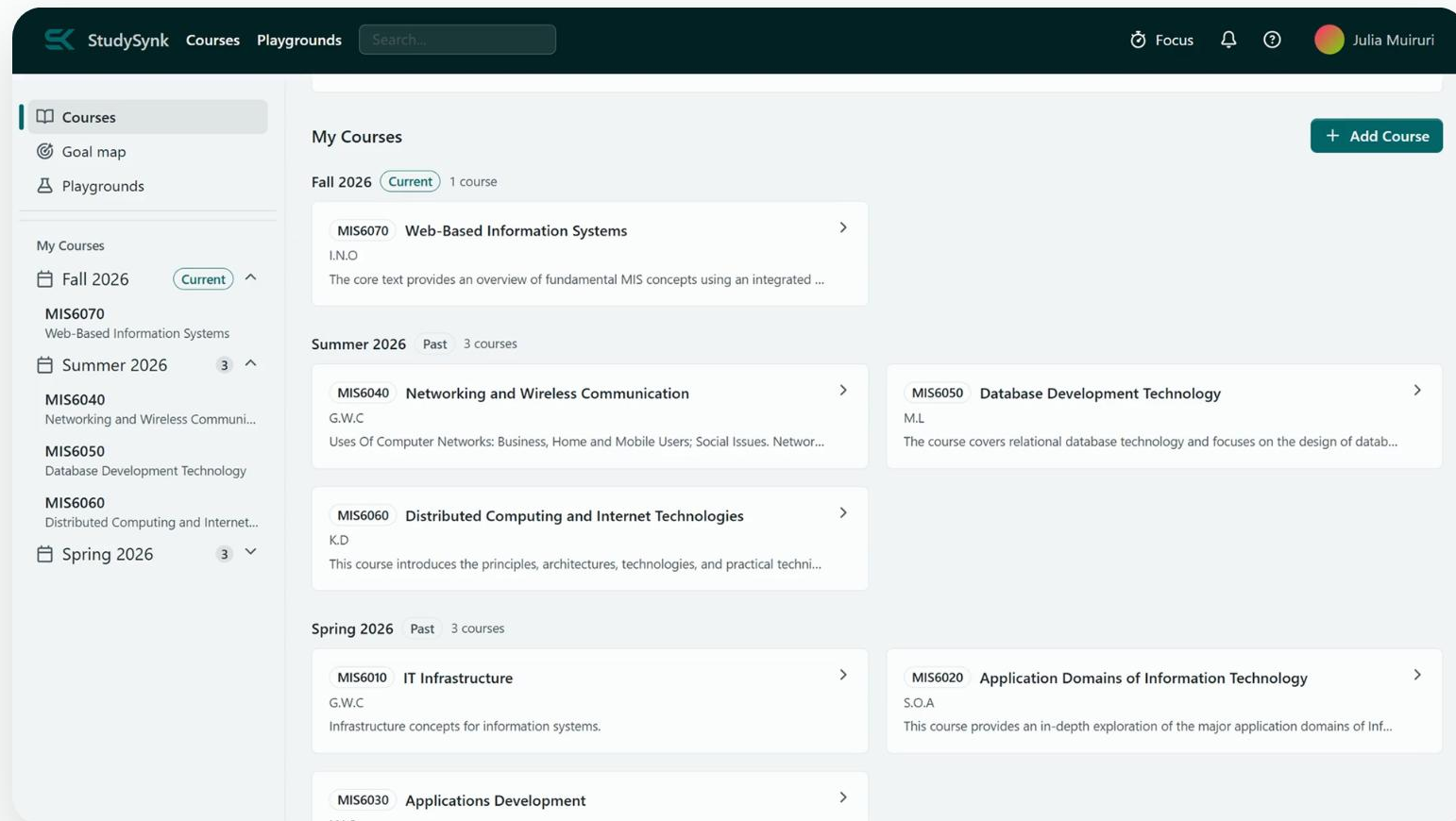
LMS, slides, PDFs, web, AI chat. The information is everywhere.



STUDYSYNK

One place

Grounded chat · answers from your materials





PROBLEM

Too much extraneous load

Abstract concepts buried in pages of text, leaving no room to think.



STUDYSYNK

Playgrounds

Active learning · extraneous ↓ germane ↑

The screenshot displays the StudySynk web application interface. At the top, a dark navigation bar contains the StudySynk logo, links for 'Courses' and 'Playgrounds', a search bar, and user controls including 'Focus', a notification bell, a help icon, and the user's name 'Julia Muiruri' next to a profile picture. The main content area is divided into a left sidebar and a right main panel. The sidebar, under the 'Courses' tab, lists 'Goal map' and 'Playgrounds'. Below this, 'My Courses' are listed by semester: 'Fall 2026' (marked 'Current'), 'MIS6070 Web-Based Information Systems', 'Summer 2026' (3 items), and 'Spring 2026' (3 items). The main panel features a light blue header with a 'USIU Africa' badge and a welcome message: 'Welcome back, Julia Muiruri! Continue your learning journey at USIU Africa.' Below this are three white summary cards: '7 Enrolled Courses', '93 Course Materials', and '7 Focus minutes · 7 days'. A light blue section titled 'Cloud security engineer' shows '5 of 7 courses connected' and an 'Open goal map' button. A white section titled 'Playgrounds' describes them as 'Interactive sandboxes for the concepts that are hard to learn from a slide deck' and includes an 'Explore playgrounds' button. At the bottom, 'My Courses' are listed again for 'Fall 2026' (Current, 1 course), showing 'MIS6070 Web-Based Information Systems' with a description: 'The core text provides an overview of fundamental MIS concepts using an integrated ...'. An '+ Add Course' button is in the top right of this section.



PROBLEM

Shrinking attention spans

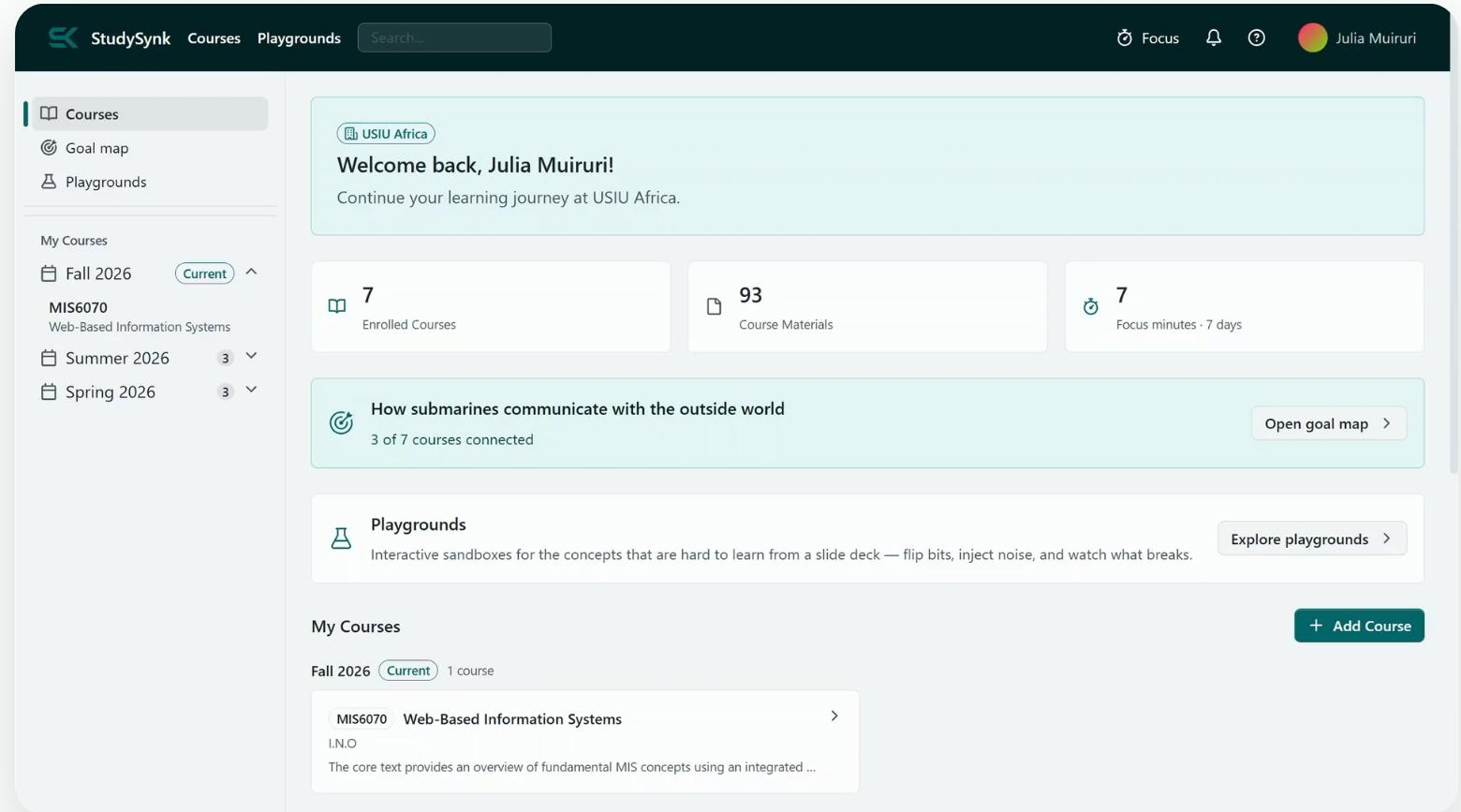
Screen attention shifts about every 47 seconds.



STUDYSYNK

Pomodoro timer

Pomodoro Technique · planned breaks





PROBLEM

Thinking in silos

Facts without connections aren't critical thinking.



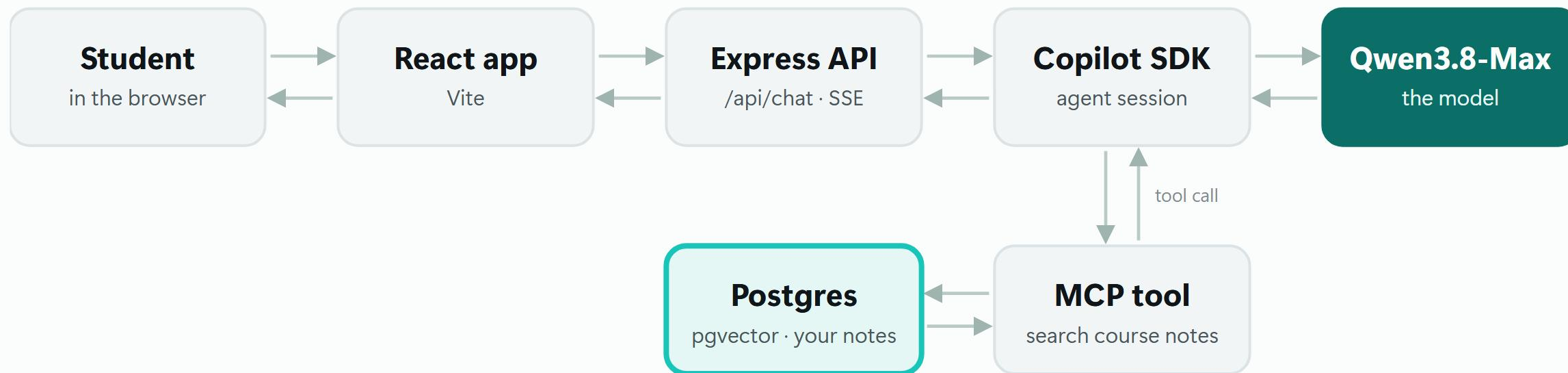
STUDYSYNK

Goal Map

Mind mapping · courses ↔ your goal

The screenshot displays the StudySynk web application interface. At the top, a dark navigation bar includes the StudySynk logo, links for 'Courses' and 'Playgrounds', a search bar, and user controls for 'Focus', notifications, help, and the user 'JuliaM'. The left sidebar features a 'Courses' menu with 'Goal map' and 'Playgrounds' options, and a 'My Courses' section listing 'Fall 2026' (Current), 'MIS6070 Web-Based Information Systems', 'Summer 2026', and 'Spring 2026'. The main content area shows a welcome message for 'JuliaM' at 'USIU Africa', followed by three summary cards: '7 Enrolled Courses', '93 Course Materials', and '7 Focus minutes · 7 days'. Below these is a section titled 'How submarines communicate with the outside world' with a subtext '3 of 7 courses connected' and an 'Open goal map' button. Further down is a 'Playgrounds' section described as 'Interactive sandboxes for the concepts that are hard to learn from a slide deck — flip bits, inject noise, and watch what breaks.' with an 'Explore playgrounds' button. At the bottom, the 'My Courses' section shows 'Fall 2026' (Current) with '1 course' listed: 'MIS6070 Web-Based Information Systems' with a brief description and a right arrow.

One question, end to end

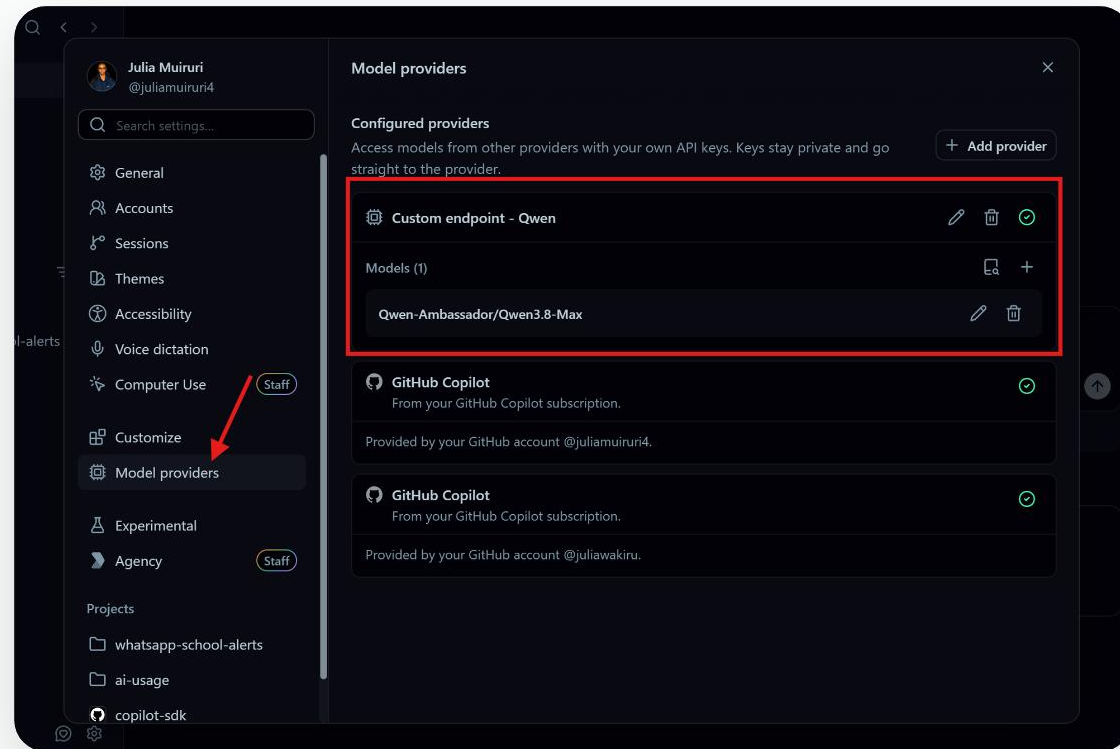


6 The MCP tool runs a vector search over your course in Postgres

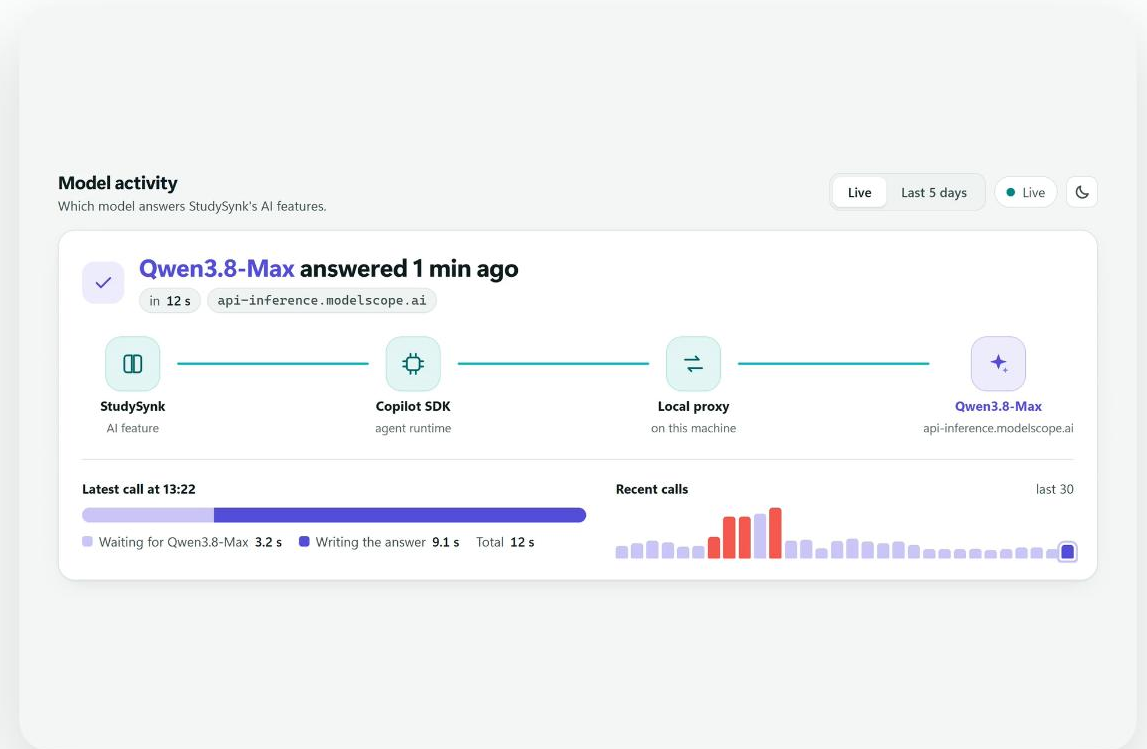
→ request ← response

HOW I BUILT IT

Built with Copilot. Powered by Qwen.



1 Qwen added as a custom model provider in the Copilot app



2 Every AI feature is answered by Qwen3.8-Max, live

GitHub Copilot app

Copilot CLI

Qwen3.8-Max, custom endpoint

Docker

Postgres + pgvector

Vitest + Jest

“

What would you rather have? A tool that **thinks for you**, or a tool that **makes you think**?

I built a tool that makes me think.

Thank you • @juliamuiruri4



Julia Muiruri

Developer Experiences (DX) at
GitHub